CST4714 Database Administration – Midterm Project Options

Your midterm project is worth **35% of your final grade**. This project is designed to give you flexibility: you may choose from **three tracks** depending on your interests, strengths, and career goals. All options are graded with the same rubric, and each will push you to apply PostgreSQL/Supabase and database administration concepts in different ways.

Option 1: Structured Database Build (Default)

This option is the most guided and is a good choice if you prefer concrete steps and hands-on SQL practice.

Tasks: - Choose a provided dataset (or propose your own, with approval). - Create a schema with **at least 3 related tables**, primary/foreign keys, and a constraint. - Load data (20+ rows per table). - Write **5 SQL queries** (joins, aggregates, subqueries, etc.). - Perform at least **one administration task** (roles/permissions, simple backup). - Submit a short report (2–3 pages) with ER diagram, queries, and reflection.

Deliverables: SQL scripts, ER diagram, short PDF report.

Option 2: White Paper / DBA Report

This option is more conceptual. It's a good choice if you enjoy **writing, research, and connecting technology to real-world problems**.

Tasks: - Identify a real-world problem from industry, science, or your own life. - Write a **3–5 page technical white paper** that includes: - A description of the problem domain. - Why PostgreSQL (or another DB type) is appropriate (or not). - A proposed schema with an ER diagram. - Discussion of DBA concerns: security, RLS, backups, scaling, DevOps. - Case study or example comparison (e.g., Supabase vs. another service).

Deliverables: PDF white paper with diagrams, citations if needed.

Option 3: Coding-Heavy DBA / DevOps Project

This option is the most technical. Choose it if you want to go deeper into **advanced SQL and database administration**.

Tasks: - Implement a schema with **4+ tables** and meaningful constraints. - Write **10+ SQL queries**, including advanced features (window functions, CTEs, recursive queries). - Create **at least two triggers** and one **view or materialized view**. - Demonstrate **roles/permissions and RLS policies** in Supabase. - Perform a **backup and restore** (document the process). - Use GitHub to store SQL scripts with a clear README.

Deliverables: SQL scripts, GitHub repo link, ER diagram, README, and a short reflection (2–3 pages).

Grading Rubric (All Tracks)

Category	Points
Schema & Design (ER diagram, constraints, normalization)	20
SQL & Analysis (queries or conceptual analysis)	25
Admin & DevOps (roles, RLS, backup, performance)	25
Documentation (report, reflection, or white paper)	20
Professionalism (organization, completeness, clarity)	10
Total	**100**

Choosing Your Option

- You must **declare your project track by Week 5**.
- If you don't choose, you'll be placed into **Option 1 (Structured Build)**.
- Regardless of your choice, you will practice SQL, database design, and core DBA skills.

Tip: Pick the track that aligns best with your goals. If you like coding → Track 3. If you like writing and problem framing → Track 2. If you want structure and a clear path → Track 1.

Due: Week 8 (see syllabus)
Submission: Upload all deliverables to Blackboard. For Track 3, include your GitHub link.

Remember: The goal of this project is not just to write SQL — it's to think like a Database Administrator. Your work should show both technical skill and an understanding of how databases are designed, managed, and applied in the real world.